

# Spectral Recovery Via Spectral-Aware Perceptual Loss

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## ABSTRACT

Hyperspectral (HS) images have applications in many scientific fields, such as remote sensing, where larger spectral resolution allows for specific detection tasks. Spectral reconstruction (SR) methods to achieve hyperspectral images from RGB or multispectral (MS) images have seen rapid and significant improvements with the use of deep convolutional networks (CNNs). While most recent works focus on developing novel architectures for SR, this work proposes a novel spectral-aware loss function component that can enhance existing learning-based methods and influence the training process to learn a more consistent spectral solution.

We perform experiments to improve existing deep learning architectures and provide evidence that the proposed loss function achieves better results when the spectral fidelity of the final image is essential. Additionally, we introduce a novel dataset, created and pre-processed from real Landsat-8/OLI and EO-1/Hyperion images, which we hope will facilitate further advances in SR. Our evidence shows that the proposed method can successfully reconstruct Hyperion (HS) spectra from OLI (MS), offering a solution for remote-sensing applications that require costly and often unavailable hyperspectral imagery. Due to anonymity concerns, the code and dataset will be made available after acceptance.

**Index Terms**—Remote Sensing, Spectral Reconstruction, Convolutional Neural Networks, Landsat8, Hyperion

## I. INTRODUCTION

Hyperspectral (HS) imaging aims to capture the reflectance of the elements in a scene along multiple spectral bands, each calibrated to be sensible on a specific wavelength. The large number of bands in an HS image permits a better approximation to the continuous spectra of the objects in contrast to multispectral (MS) imaging, which usually incorporates fewer bands. Similarly, RGB images capture only three bands. HS imaging has essential applications in many fields, such as remote sensing, climate change monitoring, agriculture, and geological exploration. These images usually provide a much better chance to accomplish these tasks than RGB or MS images because they capture the spectra at a higher resolution. HS images often present several challenges: High acquisition costs, complex noise, long pre-processing times,

| Parameters             | Landsat-8                        | EO-1            |                |
|------------------------|----------------------------------|-----------------|----------------|
|                        | OLI                              | ALI             | HYPERION       |
| Spectral Range         | 435 - 2290 [nm] (Excluding TIRS) | 480 - 2350 [nm] | 400 -2500 [nm] |
| Spatial Resolution     | 30 [m]                           | 30 [m]          | 30 [m]         |
| Band width             | 190 [Km]                         | 37 [Km]         | 7.5 [Km]       |
| Spectral Resolution    | Variable                         | Variable        | 10 [nm]        |
| Spectral Coverage      | Discrete                         | Discrete        | Continuous     |
| Pan band resolution    | 15 [m]                           | 10 [nm]         | N/A            |
| Number of bands        | 7 (Excluding TIRS)               | 10              | 220            |
| Radiometric Resolution | 12-bit                           | 12-bit          | 12-bit         |
| Temporal Resolution    | 16 days                          | 16 days         | 16 days        |

TABLE I: Technical specifications for Landsat-8 and EO-1. **Source:** eoPortal, EO-1 mission specification (EO-1 Mission Specification and Landsat-8 (L8) Data Users Handbook).

and low spatial resolution. Previous works have focused on using RGB/MS images to generate HS counterparts. This task, referred to as spectral super-resolution (SSR) or spectral reconstruction (SR), is a well-known ill-posed problem due to significant information loss when lower spectral resolution sensors measure continuous spectral scenes. Despite this, spectral characteristics and co-dependencies between different wavelengths can still be reconstructed to some extent using machine learning models.

One of the main challenges in SR applications is the lack of real-world datasets with paired MS/HS images, especially in remote sensing. This difficulty arises from the inherent challenge of simultaneously measuring a scene with multispectral and hyperspectral sensors under identical conditions.

Over time, different hyperspectral datasets have become benchmarks for validating SR methods in both indoor and outdoor scenes. For indoor scenes, datasets typically cover wavelengths from 400 nm to 700 nm. The wavelength range and number of bands vary for aerial or satellite imagery and depend on the sensor used. Regarding remote sensing, the works that have attempted to pair MS/HS images with real-world satellite data are scarce.

Regarding commonly used benchmarks for SR on hyperspectral imagery, to the best of our knowledge, none include imagery from the Hyperion sensor onboard the EO-1 satellite, which could record more than 200 wavelengths ranging from 357 nm to 2576 nm. Although the EO-1 mission ended in 2017, its historical data remains valuable as it was the only HS sensor providing open data for many years.

Like the EO-1 mission, MS remote-sensing imaging also has spectrometers widely used for scientific research. An important

example is the OLI sensor onboard the Landsat-8 mission. It was launched on February 13, 2013, and is still operational. The OLI sensor provides a wavelength range from 435 nm to 2290 nm divided into seven bands (excluding TIRS). Table I provides a more detailed comparison between Landsat-8 and EO-1 missions regarding their imaging instruments

Recently, deep learning has brought significant improvements to the SR problem using deep convolutional networks. These advancements have motivated the development of end-to-end mappings from RGB/MS to HS images. However, most methods have focused on improving the main SR architecture ignoring the valuable role of the Loss Function in a data-driven learning setup and often show limitations in capturing inter-spectral self-similarity. The performance of these methods on remote sensing and the reconstruction of HS images with hundreds of bands on real-world remote sensing imagery still remains an open problem. To address these limitations we propose a novel spectral-aware loss function component based on an autoencoder-style architecture for learning the spectral characteristics of HS images (specifically for Hyperion data). This loss function can be integrated into existing SR methods. Additionally, we introduce a new dataset constructed from Landsat-8/OLI and EO-1/Hyperion for remote sensing applications, providing a new dataset to future research on the SR problem with fully processed real-world MS and HS image pairs from these sensors.

The main contributions of this work are as follows:

- We propose a novel loss function component to better recover spectral characteristics and improve existing CNN-based methods on SR.
- We make available a new dataset for developing applications and SR methods on remote-sensing applications constructed from real-world imagery from Landsat-8/OLI and EO-1/Hyperion.
- We offer evidence of the usefulness of autoencoder architectures in learning complex characteristics on real-world images when the target is not always available at the same time as the input.
- Quantitative studies offer evidence that including a spectral-aware component in the loss function for a MS to HS learning task improves the capability of a model to learn solutions with better spectral fidelity.

The structure of this work is as follows: Section II discusses relevant works on the SR problem from a deep learning perspective. Section III explains the conformation of the Spectral-aware loss function proposed and the creation of the novel dataset OLI/Hyperion. Section IV details the experiments, metrics and datasets used to validate the proposed loss function which are discussed in V.

## II. RELATED WORKS

### A. Deep Learning Architectures for SR

With the sprung of deep-learning much of the recent work have migrated from considering the physical process of generation of a MS image from a HS ground truth to learn a direct

mapping from MS to HS images. In a learning-based setup the objective is to learn a function  $f$  such that

$$\hat{Y}_{HS} = f(X_{MS}; \theta) \quad (1)$$

is achieved by minimizing the loss

$$\arg \min_{\theta} \|\hat{Y}_{HS} - Y_{HS}\|_r, \quad (2)$$

where  $Y_{HS}$  represents the ground-truth hyperspectral image,  $\hat{Y}_{HS}$  is the reconstructed HS image,  $X_{MS}$  the MS or RGB image used for the SR task,  $\theta$  are the weight parameters of the network and  $\|\cdot\|_r$  denotes the  $l_r$ -norm, usually  $l_1$  or  $l_2$ . The goal is to learn the optimal parameters that minimize the reconstruction error for the HS image from the MS image.

Galliani et al. [1] was one of the first works to propose a Densenet-based model. Xiong et al. [2] developed a deep cnn-based network with residual connections to learn to reconstruct HS images from its spectrally undersampled counterparts. Later, Koundinya et al. [3] studied 2D and 3D convolutions for spectral reconstruction. Rangnekar et al. [4] employed a GAN framework to learn to discern between real or fake reconstructed spectral images using a Unet architecture. Gewali et al. [5] later improved the usage of a residual-based CNN model by incorporating information about the centers and spectral resolution of the MS bands along with the model to reconstruct HS spectra using the spectral response functions of the corresponding sensor.

In the later years many works have focused on developing more complex frameworks that include two or more architectures tasked to learn different aspects of the desired HS image alongside the use of transformers to leverage the use of attention along the spectral dimension of the MS. Zhao et al. [6] innovated by using pixel-shuffle in a multilevel CNN model. Li et al. [7] proposed HASIC-Net, a multi-head model with spatial and spectral sub-networks that incorporates attention to exploit inter-channel and band-wise dependencies along the used of a *Structure Information Consistency* module to maintain consistency along the SR process inside the network. Ma et al. [8] jointly achieved spatial and spectral superresolution with a model architecture that leverages the cross-attention mechanism between the spatial and spectral components of the architecture. Li et al. [9] employed multiple modules tasked with using local and global attention to capture short and long range dependencies between the spatial and spectral dimension in the generator and discriminator in a GAN framework. In a similar intuition Cai et al. [10] proposed MST++ based on Transformer to exploit the multi-head self-attention mechanism to capture these short and long range dependencies and non-local self-similarity that traditional CNNs struggle with, this work offers evidence that while HS signals are spatially sparse they tend to be spectrally self-similar, this architecture proved it's usefulness by winning the *NTIRE 2022 Spectral Reconstruction Challenge*.

### B. Commonly Used Loss Functions in SR

Although most of the work on SR have centered around innovating the architecture employed and the use of modern techniques such as attention mechanisms, the loss function of choice tends to remain relatively similar between the initial and later works with the use of L1 ([4], [6], [8]) or MSE/RMSE/MRAE ([1] - [3], [10], [11]) loss functions primarily. The use of these loss functions presents some disadvantages, for example, generating a measurement of error only on the final product of the SR model evaluated using a pixel-wise comparison while ignoring in many cases the perceptual or contextual relationships between pixels, possibly leading to solutions numerically accurate but perceptually unrealistic or even physically improbable. These loss functions also might fail to preserve fine spectral detail relying heavily on mechanisms inherent to the architecture (like an adversarial loss or attention mechanisms). Finally, neither of the previously mentioned losses capture high-level perceptual features of the spectra observed since they do not account for the visual similarity or perceptual quality, which are important in producing photo-realistic spectral reconstructions.

### C. Special Loss Functions to Enhance SR Tasks

Some works have experimented with novel loss functions aimed at capturing important characteristics of the spectra observed. For example, Gewali et al. [5] added the L2-norm of the 1st and 2nd order derivatives of the spectra to the L2-norm between the reconstruction and the ground truth, Li et al. [9] employed an adversarial framework combining pixel loss (L1-norm), a perceptual loss calculated from L1-norm of the 11th activation map of a pre-trained VGG network and finally an adversarial loss based on the idea of relativistic GAN. Finally, Li et al. [7] defined a *Spectral Gradient Loss* composed of MRAE and a constraint calculated as the mean absolute difference between the pixel values of adjacent bands. These works showcase the usefulness of the inclusion of a spectral-aware component in the loss function to improve the performance of a base architecture.

### D. Common Benchmark Datasets for SR

SR is challenging due to the inherent characteristics of HS images and the lack of real-world MS and HS datasets of the same scene under similar conditions. This is particularly difficult in remote sensing, where designing satellites with hyperspectral sensors is costly and there are few open-access sources. SR solutions are often evaluated on datasets of HS images captured indoors/outdoors or with UAVs and later degraded to simulate RGB/MS images or by selecting bands corresponding to an RGB camera. An example is the CAVE dataset [12] with 32 scenes where RGB images are created by integrating bands. A widely used benchmark is the NTIRE datasets from the *Spectral Recovery Challenge*, such as **NTIRE 2018**[13] and **ARAD1K**[14], where RGB images are taken by selecting the corresponding bands.

The previously mentioned datasets however present many difficulties when used to benchmark SR models; the disparity

between the a RGB/MS image measured in the real world with a non hyperspectral camera and the available counterparts in those datasets may be too different for a real-world scenario where a less expensive camera is not available. Another important challenge that, to the best of our knowledge, has not been sufficiently addressed is the capability to spectrally reconstruct a MS image from one sensor to simulate the HS image that another sensor would have measured, this application offers a great opportunity to facilitate research in fields where HS images are of extreme value, such as remote sensing where satellites with hyperspectral sensors are scarce and/or the target scene was not measured in the time frame desired.

He et al. [15], alongside their deep learning architecture for SR, mentions the creation of a satellite dataset comprised of MS images from the Chinese satellite GaoFen-1 and EO-1/Hyperion, to the extent of our research this work is the only recent attempt at solving the SR challenge in a real-world remote sensing setup using a dataset of MS/HS images from different sensors. Since remote sensing imagery comes in varying levels of processing ranging from the digital number (raw pixel values measured by the sensor) to radiance or reflectance values, a major deficiency in their work is the lack of specification on the processing applied to the images after measurement and thus the values that the model learned on nor is it mentioned if the pairs MS/HS were created by applying relevant filters such as cloud coverage and time difference between the two measurements, a critical aspect as measurements taken too far apart would be prone to changes on the terrain and atmospheric conditions.

## III. METHODOLOGY

### A. Motivation

In this section we propose a new spectral-aware perceptual loss function. Using an Auto-Encoder architecture instead of a pre-trained classification/regression network to recover spectrally relevant characteristics of Hyperion images and guide the training process of a CNN model to enhance the spectral resolution of MS images. We offer evidence of the validity of our approach on a dataset constructed from Landsat-8/OLI and EO-1/Hyperion images and the NTIRE 2018 dataset. Our framework is divided in two sections: The Perceptual Auto-Encoder (AE) tasked with learning the characteristics of Hyperion data and the main SR model which will be trained to increase the spectral-resolution of OLI images to match the spectra that would be observed from Hyperion.

### B. Perceptual Auto-Encoder Loss

1) *AE Architecture for Perceptual Loss*: Gatys et al. (2015) [16] proposed an innovative application based on CNN in which a previously taken photograph is transformed to match the style of a painting from an artist, for this purpose they separated their loss function into two components namely  $\mathcal{L}_{content}$  and  $\mathcal{L}_{style}$ , for the first component they extracted the intermediate activations of a pre-trained VGG-16 and compared the activation maps from the generated image and the

original photograph, for the style component they employed a similar approach but instead of comparing the raw activations they used the Gram matrix of such activations.

Let there be  $A^l$  the feature map obtained in layer  $l$ , then the Gram matrix can be defined as  $G^l \in \mathbb{R}^{N_l \times N_l}$ :

$$G^l = A^{lT} \cdot A^l. \quad (3)$$

The Gram Matrix is used to capture the style of an image, which in MS/HS images involves deeper meaning. In art, the style captures the painter's technique, but in digital images, it reflects pixel-value combinations and band correlations, defining the final color and texture. When using a specific sensor's images, the Gram matrix captures not only pixel relationships but also inter-band relations dictated by the sensor's spectral response functions. Here, the sensor acts as the artist, and its style is the spectral response functions. To address the need for a pre-trained network on Hyperion images, we use an auto-encoder that learns Hyperion image characteristics to minimize reconstruction loss.

Regarding the architecture of the Perceptual Auto-Encoder used to learn the characteristics of Hyperion Images, let  $Y_i \in \mathbb{R}^{w \times h \times B}$  the HS image, the auto-encoder presents a Unet-like architecture taking  $Y_i$  image as input. In order to make the auto-encoder learn spectrally significant features the encoder compress the image along the spectral axis only, generating a latent representation  $Z_i = \text{encoder}(Y_i)$  where  $Z_i \in \mathbb{R}^{w \times h \times b}$ , with  $b < B$ . The decoder section takes  $Z_i$  and is tasked to learn the inverse mapping of the encoder and increase the number of bands back to  $B$ , generating the final reconstruction  $\hat{Y}_i \in \mathbb{R}^{w \times h \times B}$ ,  $\hat{Y}_i = \text{decoder}(Z_i)$ .

Both the encoder and the decoder are composed of CNN blocks. The whole architecture takes a "UNet-like" shape (See Fig.1) along the spectral axis, using  $1 \times 1$  convolutions to increase the size of the spectral dimension. The architecture can be distinguished in three blocks for both the encoder and the decoder, each block formed by two convolutional layers with ELU [17] activation and between each block there is a downsampling/upsampling along the spectral axis for the encoder/decoder section respectively.

Finally, the auto-encoder is then trained with L2 loss over the Hyperion Images.

2) *Spectral-Aware Loss*: During the training phase of the main SR model the AE weights will be frozen and both the spectrally-reconstructed (from OLI) and the target (from Hyperion) images will be forwarded through the network and the Gram matrix of intermediate activations of the encoder will be used to compare the two images with  $l_2$ -norm, specifically, as shown on Fig. 1 (a) we use the activations of the last layer of each block of the encoder. The proposed perceptual loss component is then defined as:

$$\begin{aligned} \text{loss}_{\text{perceptual}} = & \|G_{UB}(Y_{\text{Hyperion}}) - G_{UB}(Y_{\text{Recon.}})\|_2 \\ & + \|G_{MB}(Y_{\text{Hyperion}}) - G_{MB}(Y_{\text{Recon.}})\|_2 \\ & + \|G_{LB}(Y_{\text{Hyperion}}) - G_{LB}(Y_{\text{Recon.}})\|_2, \end{aligned} \quad (4)$$

where  $G_l$  is the Gram matrix of the reshaped activations of the  $l$ -block,  $Y_{\text{Hyperion}}$  is the target Hyperion image and  $Y_{\text{Recon.}}$  the spectrally-reconstructed image from OLI.

In order to guarantee that the spatial characteristics of the image are also taken into account by the main SR model a content component is also included in the total loss function, in this case we used the  $l_2$ -norm between the reconstructed  $Y_{\text{Recon.}}$  image and the target  $Y_{\text{Hyperion}}$ :

$$\mathcal{L}_{\text{content}} = \|Y_{\text{Hyperion}} - Y_{\text{Recon.}}\|_2. \quad (5)$$

The final spectral-aware loss function is then defined as

$$\mathcal{L}_{\text{final}} = \alpha \mathcal{L}_{\text{content}} + \beta \mathcal{L}_{\text{perceptual}}, \quad (6)$$

where  $\alpha$  and  $\beta$  are hyperparameters that control the influence of the content and perceptual components.

### C. SR Architecture

For this study we used the same UNet-style architecture previously employed for the Perceptual Auto-Encoder as the main backbone for testing the proposed loss function from equation 6, that is convolutional blocks with ELU activation and two skip connections between the Upper Block and Middle Block. The main reason is that this architecture is clean from other methods that can obscure the benefit of the proposed loss function such as attention mechanisms.

We compare the results with the architecture employed by koundinya et al. (2018)[3] (Only of convolutional layers with ReLU activation) since it aligns with the paradigm of not including other mechanisms to specifically enhance the spectral resolution.

### D. OLI-Hyperion Dataset

The Worldwide Reference System-2 (WRS-2) is a notation for referencing the position of LandSat satellites over the Earth's surface, where data is available. This notation divides the Earth's surface into  $(\text{path}, \text{row})$  tuples: paths range from 001 to 251, denoting orbital tracks numbered east to west, starting at the equator crossing at  $65.48^\circ$ . These paths are approximate and require periodic adjustments due to satellite drift. Rows refer to the latitudinal center line of a frame of imagery. WRS-2 is used by LandSat missions 4, 5, 7, 8, and 9, as well as EO-1/Hyperion, which was launched in formation with LandSat-7 in 2000. EO-1/Hyperion images are paired with LandSat-8/OLI (launched in 2013) to utilize its advanced sensor technology, as the LandSat-8/OLI sensor is newer than LandSat-7's.

A dataset composed from images from LandSat-8/OLI and EO-1/Hyperion sensors was constructed with the following restrictions:

- 1) The dataset must contain tuples  $(x_i, y_i)$  from the same scene measured both from LandSat8/OLI ( $x_i$ ) and EO1/Hyperion ( $y_i$ ) to allow for a supervised learning setup. This means that  $x_i$  and  $y_i$  belong to the same tuple  $(\text{path}, \text{row})$  or the corresponding areas between both images superpose at some point on the surface.

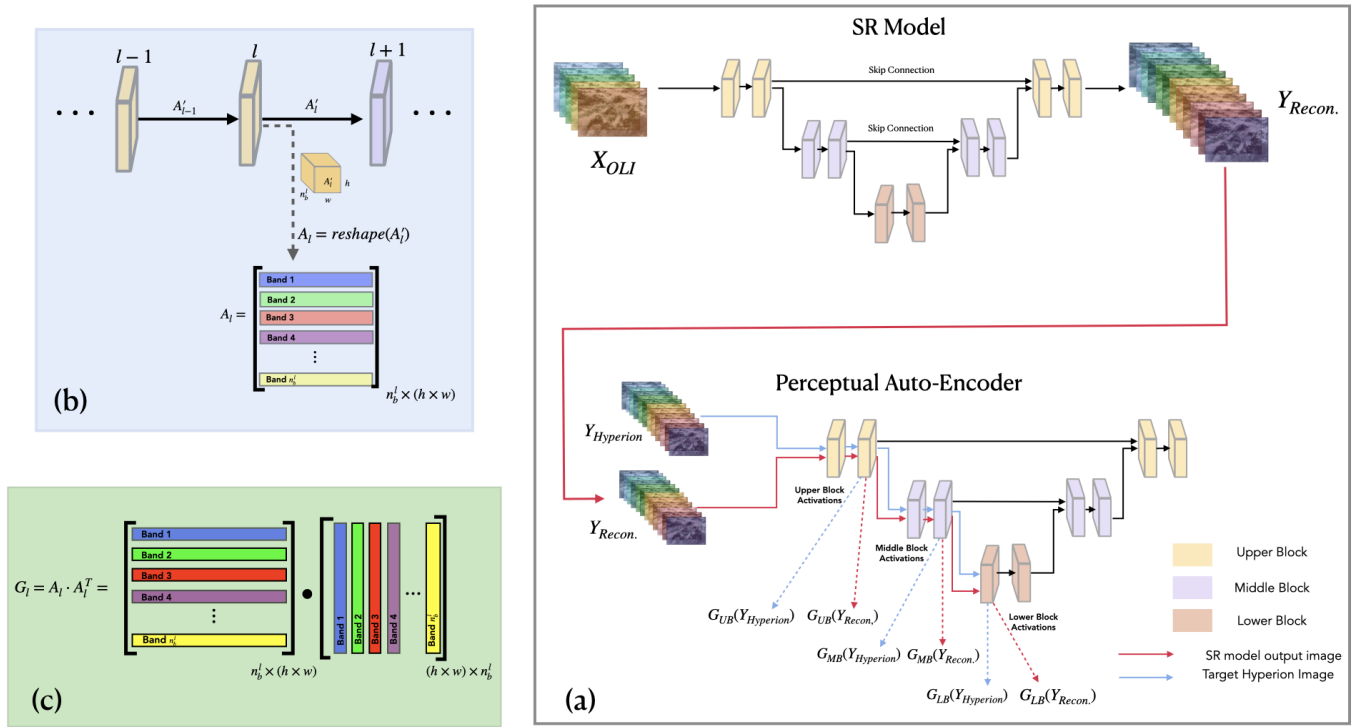


Fig. 1: **(a)** shows the complete framework proposed in this work, as an example for the OLI/Hyperion dataset, once the perceptual auto-encoder is trained using only HS images, the training of the SR model is made by making a forward pass using a MS image ( $X_{OLI}$ ) through the SR model and the reconstructed HS image ( $Y_{Recon.}$ ) is then forwarded through the Perceptual Auto-Encoder alongside the HS Hyperion image, the activation maps of three blocks in the encoder are extracted ( $A'$ ) and reshaped as shown in **(b)** obtaining  $A$ , finally for each activation extracted the corresponding Gram Matrices (namely  $G_{UB}(\cdot)$ ,  $G_{MB}(\cdot)$ ,  $G_{LB}(\cdot)$ ) is calculated as shown in **(c)** which in turn are used to calculate the perceptual component of the loss function, showed in equation 6.

- 2) The temporal difference between the measurement of a scene from both sensors must be the least possible and no greater than a month, this is for ensuring that the climate effects on the terrain would be about the same.
- 3) The scene proportion covered by clouds must be less than 10% for both images. This was deemed necessary in order to minimize the discrepancies between the images not because a change on the spectra of the components on the surface, but because occlusion from the clouds, a problem that passive sensors such as OLI and Hyperion are sensible.

#### 1) Pre-Processing OLI and Hyperion Data Procedure:

As data products from both EO-1 and LandSat-8 come in varying levels of processing we choose the highest available level for the images that matched up the criteria previously mentioned, for OLI images we used level *L1TP* (Precision and Terrain Correction, the highest level available for Landsat-8 data products) as for Hyperion we used level *L1T* (Radiometric and systematic geometric corrections). For LandSat-8 *L1TP* data we applied radiometric correction following the LandSat-8 Users Handbook<sup>1</sup> recommendations. Finally, having the images from both sensors converted to radiance values (product of the radiometric calibration procedure) we

applied atmospheric corrections according to the individual characteristics of the scenes using the FLAASH<sup>2</sup> tool on the ENVI software to convert all pixel values of the images from radiance values to reflectance values. The complete parameters of each processing operation is made available along with versions of the images in their original format, post-radiometric calibration and post-atmospheric calibration as part of the dataset in case a future work requires them.

Through the time of service of the Hyperion sensor many bands have showed problems and are usually discarded since they are unusable to get a proper reflectance measurement [18]. We removed such bands reducing the total bands in the Hyperion HS images to 175.

Finally, after extracting a subset of images from OLI and Hyperion with the same path/row and process them to reflectance values we took the intersection of both images to obtain the MS and HS pairs. In order to better manage the size of the images after obtaining the intersection of the products we divided each intersection on tiles of  $64 \times 64$  pixels which were used to train the SR model. A summary of the resulting OLI/Hyperion dataset can be seen on Table II.

<sup>1</sup><https://www.usgs.gov/landsat-missions/landsat-8-data-users-handbook>

<sup>2</sup><https://www.nv5geospatialsoftware.com/docs/FLAASH.html>

| Subset       | N° Scenes | N° Examples |
|--------------|-----------|-------------|
| Train        | 5         | 2151        |
| Validation   | 1         | 540         |
| Test         | 3         | 1953        |
| <b>Total</b> |           | <b>4644</b> |

TABLE II: Specifications of the OLI/Hyperion dataset constructed for this work.

#### IV. EXPERIMENTS

##### A. Datasets

We employ two different datasets to evaluate the proposed spectral-aware perceptual loss, the first one is the dataset also proposed in this work: OLI/Hyperion. As the second one we choose the dataset used for the challenge on spectral recovery for the 2018 NTIRE Competition [13] where the aim of the challenge was to reconstruct HS spectra from indoor and outdoor scenes using the RGB version of the HS images. Table III shows information about both datasets.

| Dataset      | N° Bands MS | N° Bands HS | N° Train Samples | N° Validation Samples | N° Test Samples |
|--------------|-------------|-------------|------------------|-----------------------|-----------------|
| OLI/Hyperion | 7           | 175         | 2151             | 540                   | 1953            |
| NTIRE2018    | 3           | 31          | 204              | 52                    | 5               |

TABLE III: General Information from the two datasets used to evaluate the proposed loss function.

##### B. Evaluation Metrics

We used four widely employed evaluation metrics for the SR task: MSE, RMSE, SSIM and SAM. Additionally, since the analysis of the pixel values in a HS image is very relevant to the remote sensing context as a fully processed image (as the ones from the OLI/Hyperion dataset) represent reflectance values of the observed scene we calculated the RMSE scaled to reflectance to quantify in a physical manner the errors.

##### C. Experimental Details

1) *OLI/Hyperion Dataset*: The train process for the auto-encoder tasked with learning was optimized with Adam [19] with  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$  and learning rate initialized at 0.0001. The layers of the encoder section were implemented in a way that the spectral dimension of the activation maps followed a sequence of 175/90/45/10 to the final section of the encoder, the decoder was mirrored using this configuration following an increasingly spectral dimension to finally generate the 175 bands of the Hyperion ground truth.

For the SR architecture, as mentioned before, we employed the same architecture used for the auto-encoder following the same activation maps pattern (175/90/45/10).

2) *NTIRE 2018 Dataset*: For the NTIRE dataset the auto-encoder, the encoder was designed to follow a pattern of 31/25/20/15 and a mirrored decoder. The SR architecture used a pattern of 25/20/15 (mirrored decoder).

We trained all SR models using two loss functions: MSE and MSE + Perceptual in order to show the benefits of including a perceptual component to a base loss function. For the Perceptual component proposed in equation 4 we used  $\alpha = 1$  and  $\beta = 0.001$ . Both perceptual auto-encoders were trained using 80% of the HS data.

We implemented all models with the PyTorch and train them on a NVIDIA RTX 2070 GPU. Table III shows the train-test split used for the experiments.

#### V. RESULTS

1) *General Results for the SR task*: We compared the performance of the perceptual component on both datasets from III, the results for the OLI/Hyperion dataset are listed in Table IV using the two different architectures used for the SR task along with the loss function used to train the model.

The performance of trained architectures on the OLI/Hyperion dataset (Table IV) shows that Unet-based architectures outperform simple CNNs in reflectance error, SSIM, and SAM metrics. Training with the proposed perceptual loss function also improves performance over the classic MSE criterion. For example, a simple CNN trained with the perceptual loss shows a 0.0053 decrease in reflectance error (7.27% improvement), a 0.061 increase in SSIM (17.1% improvement), and a slight decrease in the spectral angle (9.10% improvement) compared to training with MSE.

By looking at the spectral reconstruction curves for both variants of the UNet-based SR architectures (figures 4 and 5), a small improvement in the spectrum prediction is observed visually. For example, for all the reconstructions shown of the model trained with MSE, a general underestimation of the spectrum value is observed for wavelengths close to  $0.5 [\mu m]$  and in the case of the third scene shows a significant difference between the real and predicted spectrum.

Regarding the errors per groups of bands (figures VII and VI) a clear improvement in the estimated reflectance value can be observed when using the perceptual loss function and the UNet architecture (5.75% decrease improvement) over the full test dataset. Improvements are also observed in the value of SSIM (1.46% increase) and SAM (3.43% decrease). A special decrease in the value of the error in reflectance is observed for the SWIR 1 and SWIR 2 bands (11.41% and 16.60% improvement respectively).

Table V shows the results for the NTIRE 2018 dataset. For this dataset the results of the trained SR Unet-like model show considerable improvements with respect to the RMSE when using a perceptual loss function but not for the SSIM and SAM which remain very similar in both variants of the loss function, being in any case slightly lower for the architecture trained with perceptual function.

| Architecture | Loss Function    | Perceptual Architecture | MSE        | RMSE    | RMSE scaled to reflectance | SSIM    | SAM     |
|--------------|------------------|-------------------------|------------|---------|----------------------------|---------|---------|
| CNN          | MSE              | -                       | 537094.030 | 732.866 | 0.07328                    | 0.36121 | 0.23311 |
|              | MSE + Perceptual | AE 90/45/10             | 461736.502 | 679.511 | 0.06795                    | 0.42282 | 0.21195 |
| UNet-like    | MSE              | -                       | 397764.405 | 630.685 | 0.06306                    | 0.49794 | 0.19773 |
|              | MSE + Perceptual | AE 90/45/10             | 353297.060 | 594.387 | 0.05943                    | 0.50520 | 0.19094 |

TABLE IV: Experimental results for the OLI/Hyperion dataset.

| Architecture | Loss Function    | Perceptual Architecture | MSE       | RMSE    | SSIM    | SAM     |
|--------------|------------------|-------------------------|-----------|---------|---------|---------|
| CNN          | MSE              | -                       | 8491.696  | 92.150  | 0.68656 | 0.15431 |
|              | MSE + Perceptual | AE 25/20/15             | 5840.583  | 76.423  | 0.69594 | 0.12878 |
| UNet-like    | MSE              | -                       | 17952.693 | 133.987 | 0.598   | 0.14699 |
|              | MSE + Perceptual | AE 25/20/15             | 7557.366  | 86.933  | 0.577   | 0.14429 |

TABLE V: Experimental Results obtained from the architectures used for SR on the NTIRE 2018 dataset.

| Band Group      | MSE        | RMSE    | RMSE scaled to reflectance | SSIM    | SAM     |
|-----------------|------------|---------|----------------------------|---------|---------|
| Full Test Set   | 353297.060 | 594.388 | 0.05943                    | 0.50520 | 0.19094 |
| Coastal Aerosol | 313264.090 | 559.700 | 0.05597                    | -       | -       |
| Blue            | 235322.010 | 485.100 | 0.04851                    | -       | -       |
| Green           | 141278.874 | 375.871 | 0.03759                    | -       | -       |
| Red             | 98634.383  | 314.061 | 0.03141                    | -       | -       |
| NIR             | 569439.933 | 754.612 | 0.07546                    | -       | -       |
| SWIR 1          | 197303.407 | 444.188 | 0.04442                    | -       | -       |
| SWIR 2          | 152544.861 | 390.570 | 0.03906                    | -       | -       |
| Pan             | 102088.668 | 319.513 | 0.03195                    | -       | -       |

TABLE VI: Experimental results for the architecture for spectral SR based on UNet with 175/90/45/10 configuration and trained with perceptual loss function, using a 175/90/45/10 perceptual architecture, on the test set of the OLIHyper dataset generated in this work.

| Band Group      | MSE        | RMSE    | RMSE scaled to reflectance | SSIM    | SAM     |
|-----------------|------------|---------|----------------------------|---------|---------|
| Full Test Set   | 397764.405 | 630.685 | 0.06306                    | 0.49794 | 0.19773 |
| Coastal Aerosol | 664388.010 | 815.100 | 0.08151                    | -       | -       |
| Blue            | 504821.377 | 710.508 | 0.07105                    | -       | -       |
| Green           | 226505.627 | 475.926 | 0.04759                    | -       | -       |
| Red             | 143234.795 | 378.464 | 0.03785                    | -       | -       |
| NIR             | 594920.034 | 771.311 | 0.07713                    | -       | -       |
| SWIR 1          | 251445.835 | 501.444 | 0.05014                    | -       | -       |
| SWIR 2          | 219261.468 | 468.254 | 0.04683                    | -       | -       |
| Pan             | 202737.787 | 450.264 | 0.04503                    | -       | -       |

TABLE VII: Experimental results by band group for the architecture for spectral SR from a simple CNN and with L2 loss function on the test set of the OLIHyper dataset.

2) *Visual results on reconstructed spectra for OLI/Hyperion Dataset*: Finally, figures 2 - 5 show visual reconstructions for one random pixel selected from a random tile on the test split for the OLI/Hyperion dataset.

For each figure on the left, three pairs of images randomly selected from the test set are shown, corresponding to the image reconstructed by the architecture (top image) and the target image from Hyperion. In addition, graphs of the spectrum reconstructed by the architecture are shown on the left. of SR (orange line), the target spectrum to be reconstructed (blue line) and the spectrum delivered to the model as input image from the Landsat8 OLI sensor (green line). The spectra mentioned above were extracted from a random pixel of the scene that can be seen in the images on the left, marked as the intersection of the two red lines.

## VI. CONCLUSIONS

In this paper we propose, to the best of our knowledge, the first loss function to capture spectral characteristics of a sensor and utilize it to train a backbone model by using the Gram matrix of the activation maps of an autoencoder trained for reconstructing the target HS images, furthermore, we propose a new dataset that pairs real-world MS/HS satellite imagery from different missions/sensors and offer evidence that it can provide a useful tool to analyze up-to-date MS imagery from an active mission in an hyperspectral manner, quantitative and qualitative analysis of the reconstruction of Hyperion images from OLI show that in many cases even a simple model can learn an effective mapping between the two sensors. Lastly, we believe that the dataset OLI/Hyperion and the Perceptual Loss for spectral characteristics open new possibilities for scientific applications that rely heavily on remote-sensing HS images that are unavailable or restrictive due to the high costs of acquisition of such images.

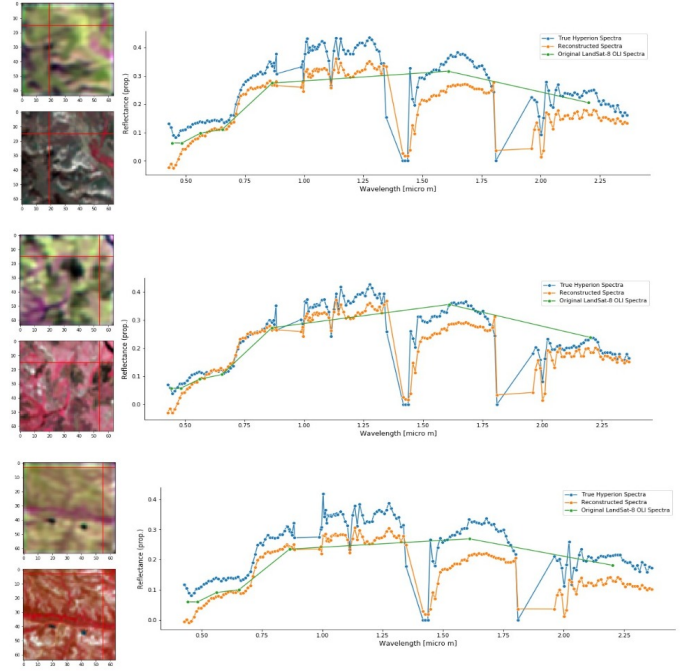


Fig. 2: Experimental results for spectral reconstruction using the architecture based on Koundinya et al., with MSE loss function.

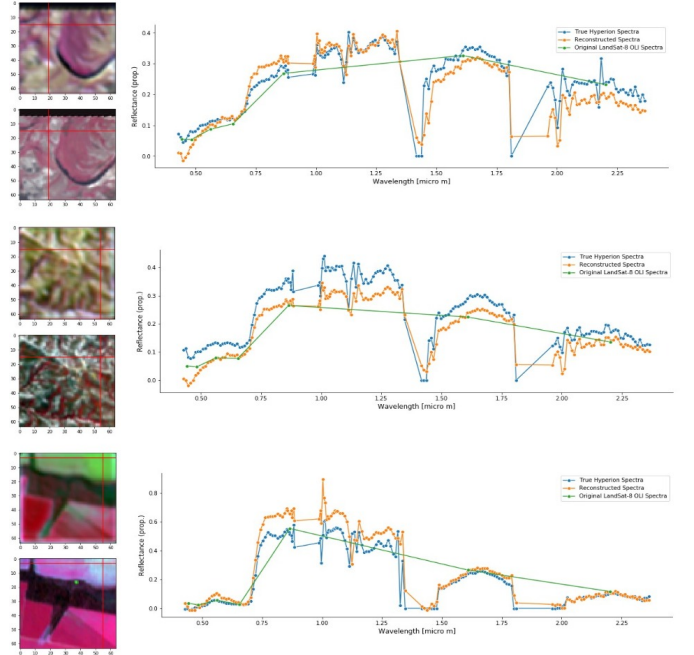


Fig. 3: Experimental results for SR using the architecture based on Koundinya et al., with perceptual loss function using an AE architecture with 90/45/10 layer configuration.

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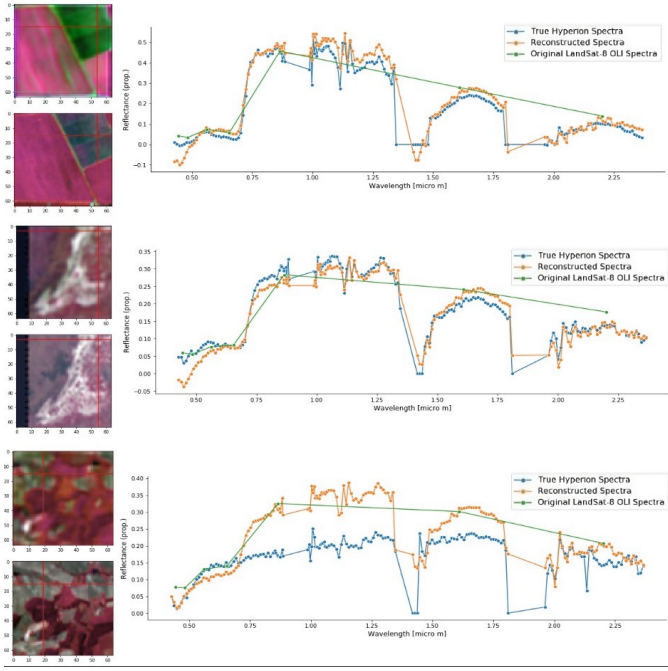


Fig. 4: Experimental results for spectral reconstruction using the UNet-based architecture with 175/90/45/10 configuration and with MSE loss function.

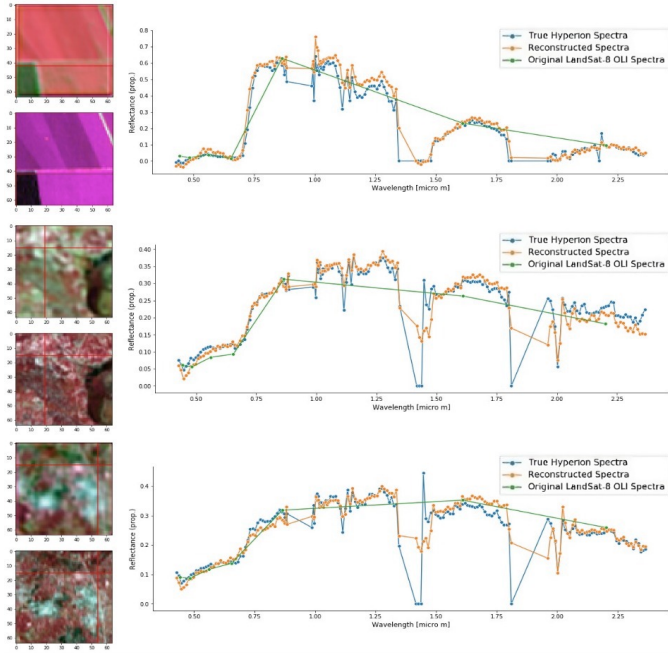


Fig. 5: Experimental results for SR using the UNet-based architecture with 175/90/45/10 configuration and with perceptual loss function, using a perceptual architecture with 175/90/45/10 configuration.

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